

AI Powered Debt Relief & Financial Recovery Platform

Project Overview

The AI Powered Debt Relief & Financial Recovery Platform is an intelligent web application designed to simplify the debt management and financial recovery process for borrowers. The platform leverages Artificial Intelligence to analyze a user's financial condition, predict suitable settlement amounts, and automatically generate professional negotiation letters for lenders.

The system provides borrowers with a centralized dashboard where they can monitor loan information, evaluate financial health, and receive AI-assisted financial recommendations. By integrating Google Gemini AI with modern web technologies, the platform helps borrowers make informed financial decisions while reducing manual effort in settlement negotiations.

Problem Statement

Many borrowers struggle to negotiate loan settlements due to limited financial knowledge, lack of proper communication with lenders, and the absence of intelligent financial guidance. Traditional debt settlement processes are time-consuming and require manual preparation of negotiation letters.

This platform addresses these challenges by automating financial analysis and providing AI-generated settlement recommendations and negotiation letters.

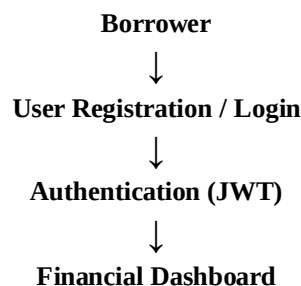
Objectives

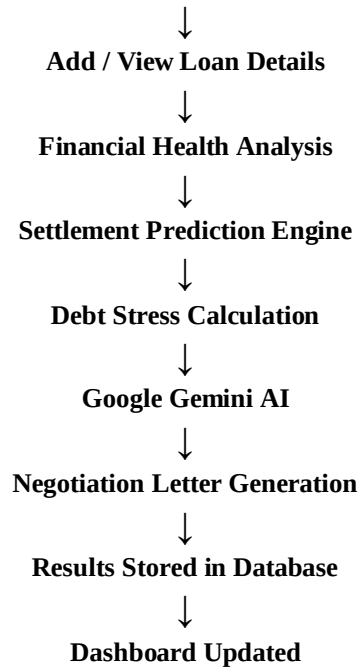
The major objectives of the project are:

- Provide secure borrower authentication.
- Maintain borrower loan records.
- Analyze financial health.
- Calculate debt stress.
- Predict optimal settlement percentages.
- Generate AI-powered negotiation letters.
- Track previous AI recommendations.
- Improve borrower decision-making using AI.

System Workflow

The complete workflow of the application is shown below.





Detailed Workflow

Step 1: User Authentication

The borrower registers a new account or logs into the system using their email and password. FastAPI validates the credentials and generates a secure JWT token, allowing only authenticated users to access the platform.

Input:

- Email
- Password

Output:

- Secure Login
- JWT Token

Step 2: Financial Dashboard

After successful login, the borrower is redirected to the dashboard where all financial information is displayed.

The dashboard includes:

- Monthly Surplus
- EMI Ratio
- Debt Stress Level
- Loan List
- Settlement Prediction
- Negotiation Letter Generation

This dashboard acts as the central control panel for the borrower.

Step 3: Loan Management

The borrower enters loan information such as:

- Bank Name
- Outstanding Loan Amount
- EMI Amount
- Overdue Duration
- Monthly Income
- Monthly Expenses

These records are stored securely in the SQLite database using SQLAlchemy ORM.

Step 4: Financial Health Analysis

The Financial Health Engine processes borrower information to determine financial stability.

The engine calculates:

- Monthly Surplus
- Monthly Deficit
- EMI Ratio
- Debt-to-Income Ratio
- Financial Stress Score

These metrics help determine the borrower's repayment capability.

Step 5: Debt Stress Analysis

The Debt Stress Analyzer evaluates the borrower's financial burden by comparing loan obligations against income.

The system classifies borrowers into categories such as:

- Low Risk
- Moderate Risk
- High Risk

This classification assists in predicting an appropriate settlement amount.

Step 6: AI Settlement Prediction

Once financial analysis is complete, the Settlement Prediction Engine estimates a suitable settlement recommendation.

The prediction includes:

- Settlement Percentage
- Estimated Settlement Amount
- Debt Stress Score
- Financial Health Status

The recommendation is generated based on predefined financial rules and borrower affordability.

Step 7: AI Negotiation Letter Generation

The borrower selects a loan and requests an AI-generated negotiation letter.

The backend sends financial information to the Google Gemini API.

The AI generates:

- Professional settlement request letter
- Financial hardship explanation
- Negotiation email
- Lender-specific settlement proposal

The borrower can review and use the generated letter for communication with the lender.

Step 8: History Management

Every AI-generated recommendation and negotiation letter is stored in the database.

Users can access:

- Previous settlement predictions
- AI-generated letters
- Negotiation history
- Financial reports

This allows borrowers to monitor progress over time.

System Modules

User Authentication Module

Responsibilities:

- Registration
- Login
- Password Verification
- JWT Authentication

Dashboard Module

Displays:

- Financial Overview
- Active Loans
- Monthly Surplus
- EMI Ratio
- Stress Level

Loan Management Module

Responsible for:

- Adding Loans
- Updating Loan Information
- Viewing Loans
- Loan History

Financial Health Module

Calculates:

- Monthly Income
- Monthly Expenses
- EMI Ratio
- Debt Stress
- Financial Stability

Settlement Prediction Module

Predicts:

- Settlement Percentage
- Settlement Amount
- Risk Score
- Financial Recommendation

AI Negotiation Module

Uses Google Gemini AI to generate:

- Settlement Request Letter
- Negotiation Email
- Professional Financial Explanation

Database Module

Stores:

- User Information
- Loan Records
- Financial Metrics
- Settlement History
- AI Negotiation Letters

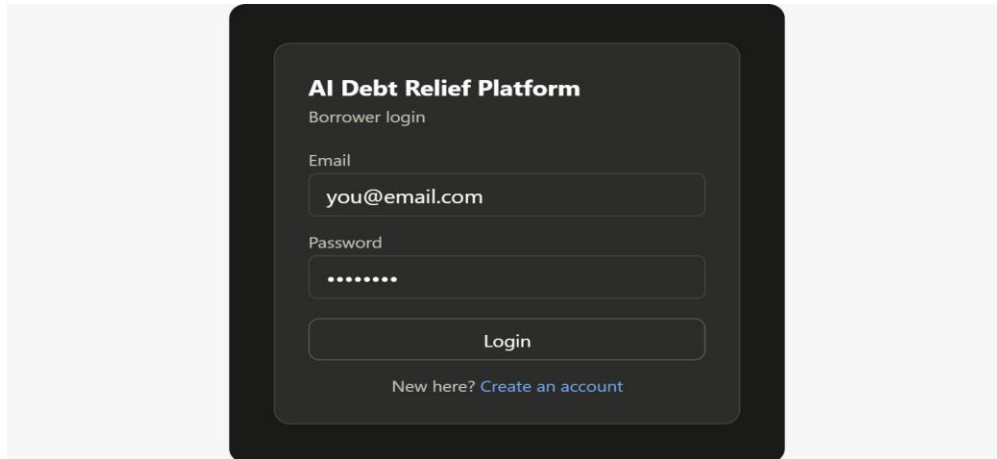
Technology Stack

Component	Technology
Frontend	React.js + Vite
Backend	FastAPI
Language	Python
Database	SQLite
ORM	SQLAlchemy
AI Model	Google Gemini API
Authentication	JWT
API Format	REST API
Version Control	Git & GitHub

Application Screens

Login Page

The login page authenticates borrowers using secure credentials before granting access to the application.

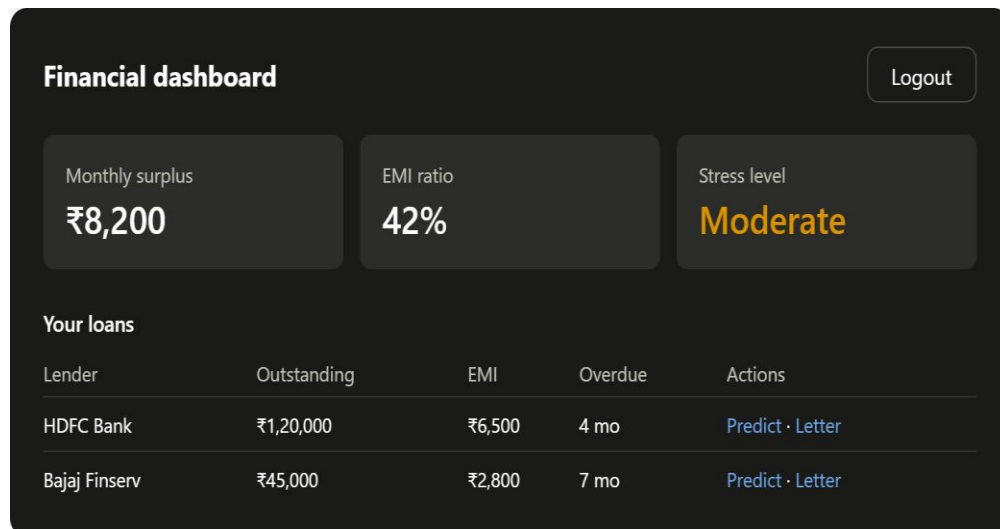


The login form is titled "AI Debt Relief Platform" and "Borrower login". It contains fields for "Email" (with the placeholder "you@email.com") and "Password" (with masked characters "....."). A "Login" button is positioned below the password field. At the bottom, there is a link that says "New here? [Create an account](#)".

Login Page

Financial Dashboard

The dashboard presents key financial metrics including monthly surplus, EMI ratio, debt stress level, and active loan details. Users can also access settlement prediction and negotiation features from this page.



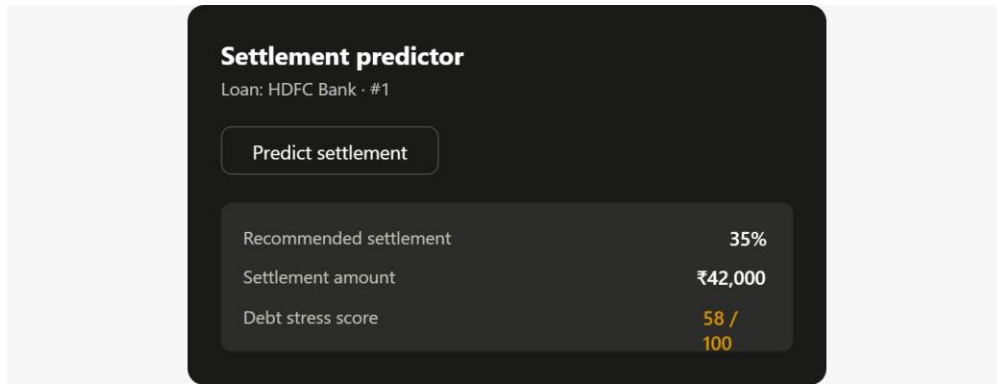
The financial dashboard displays three key metrics: "Monthly surplus" at ₹8,200, "EMI ratio" at 42%, and "Stress level" as Moderate. Below these metrics is a section titled "Your loans" containing a table of active loans.

Lender	Outstanding	EMI	Overdue	Actions
HDFC Bank	₹1,20,000	₹6,500	4 mo	Predict · Letter
Bajaj Finserv	₹45,000	₹2,800	7 mo	Predict · Letter

Financial Dashboard

Settlement Predictor

The settlement prediction module calculates the recommended settlement percentage, settlement amount, and debt stress score based on the borrower's financial profile.

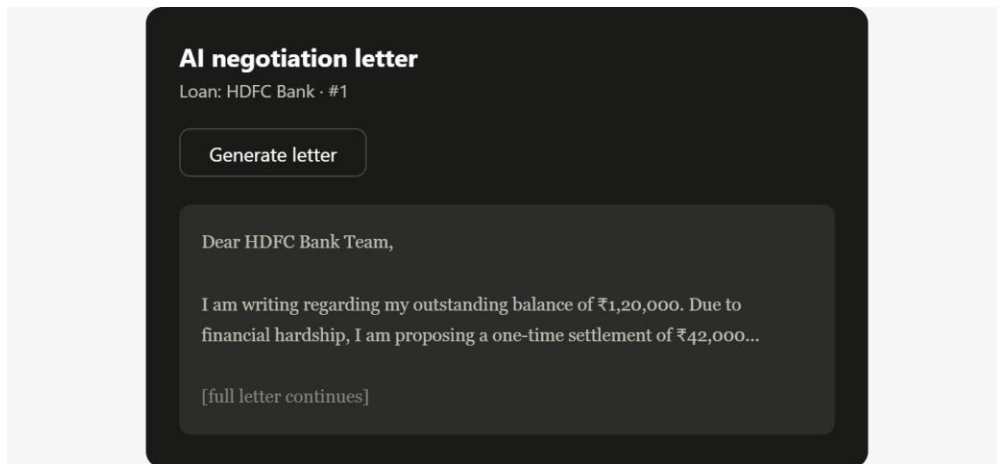
A dark-themed user interface for a settlement predictor. At the top, it says "Settlement predictor" in bold, followed by "Loan: HDFC Bank · #1". Below this is a button labeled "Predict settlement". Underneath the button is a table with three rows: "Recommended settlement" with a value of "35%", "Settlement amount" with a value of "₹42,000", and "Debt stress score" with a value of "58 / 100".

Recommended settlement	35%
Settlement amount	₹42,000
Debt stress score	58 / 100

Settlement Predictor

AI Negotiation Letter

The negotiation module generates professional lender-specific settlement request letters using Google Gemini AI based on the borrower's repayment capacity.

A dark-themed user interface for an AI negotiation letter generator. At the top, it says "AI negotiation letter" in bold, followed by "Loan: HDFC Bank · #1". Below this is a button labeled "Generate letter". Underneath the button is a text area containing a sample letter. The letter starts with "Dear HDFC Bank Team," followed by a paragraph: "I am writing regarding my outstanding balance of ₹1,20,000. Due to financial hardship, I am proposing a one-time settlement of ₹42,000...". It ends with "[full letter continues]".

Dear HDFC Bank Team,

I am writing regarding my outstanding balance of ₹1,20,000. Due to financial hardship, I am proposing a one-time settlement of ₹42,000...

[full letter continues]

AI Negotiation Letter

Advantages

- AI-assisted financial decision making
- Automated settlement recommendations
- Professional negotiation letter generation
- Secure borrower authentication
- User-friendly dashboard
- Faster debt analysis
- Scalable architecture
- Reduced manual effort

Future Enhancements

- Multi-bank integration
- Credit score prediction

- OCR-based loan document upload
- Email and SMS notifications
- Cloud deployment
- Multi-language support
- Voice-enabled AI assistant
- Machine Learning-based settlement prediction
- Mobile application support

Conclusion

The AI Powered Debt Relief & Financial Recovery Platform provides an intelligent and efficient solution for borrowers seeking financial recovery. By integrating AI-powered financial analysis with automated negotiation support, the platform simplifies debt management and improves communication between borrowers and lenders. Its modular architecture, secure authentication, and scalable design make it a practical foundation for future financial technology applications.